Name : AKASH SAMPATH

TITLE : Research_Q_A_Part_1

Q1) What is HTTP Protocol ?

- HTTP is basically stands for Hypertext Transfer Protocol
- A systems that allows communication between browsers like Google Chrome and other websites
- When we visit a website , the browser uses HTTP to send a request to the server hosting that site and server sends back the data needed to display the page
- Also set of rules that lets your browser and web server communicate ensuring websites load correctly
- A protocol fetches resources such as HTML documents and also a foundation of World Wide Web (www) used to load web pages
- It govern how information will be transferred between network devices , specifically web servers and client browsers
- HTTP was invented alongside HTML to create first interactive text based web browser
- Follows classical client-server model with client opening a connection to make request then waiting until it receives a response from the server
- This is stateless protocol where the server does not keep any session data between two requests where cookies also adds state to interact with client and server
- HTTP is an application layer protocol used by web browsers to transfer text , images and multimedia across distributed web systems
- Enables the transfer of data over the internet allowing users to access websites and other online resources

Working of HTTP :

Open Web Browser :

- Typing a website URL : www.example.com

DNS Lookup :

- Browser asks a Domain Name System known as DNS Server to find out Ip address associated with the URL
- Generic Domain are like .com , .org , .net which Is top level DNS and widely used
- Country Code DNS are like .in , .us represents specific countries or regions

Send HTTP Request :

- Once the browser has the website IP address
- It sends an HTTP request to the server
- The request asks the server for the resources needed to display the page like text , images , videos

Server Response :

- Server process your request and sends back as HTTP response
- Response contains the requested resources like HTML , CSS , JavaScript needed to load the page

Rendering the Web Page :

- Browser receives the data from the server and displays the webpage on our screen

Note : Therefore after the page is loaded , the connection between the browser and server is closed , if you request a new page , a new connection will be made

Q2) What is Web Application ?

- A web application is a software program that runs on the remote server and is accessed through a web browser over the internet
- Unlike traditional applications that require installation on your device , web application directly work from browser such as Google Chrome and Firefox
- Web applications need not to be download or installed , users can access them directly via the internet using any device
- Web applications rely on backend server , frontend user interface database , API and browser
- These are main components which support the working on an web application

Forms of Web Application :

Static Web Application :

- It delivers fixed content
- Typically used for portfolios

Dynamic Web Application :

- It responds to user input and generates real time content
- Such as social media platforms and collaborative tools like Google Docs

Ecommerce Web Applications :

- It enables online shopping
- By securing transactions like Amazon and Flipkart

Content Management System :

- Known as CMS
- Allows users to manage and publish content without coding such as Word Press

Benefits of Web Application :

Cross Platform Compatibility :

- Helps to work seamlessly across different operating system and devices through any modern web browsers

No Installation Required :

- The users can access application immediately without downloading or installing software

Automatic Updates :

- The update are deployed automatically on the server side ensuring all users have the latest features and security patches

Cost Effective Development :

- A single web application serves all platform than developing multiple applications

Enhanced Security :

- Security is immediately deployed by updating patches by upgradation by security management team

Scalability :

- Can easily accommodate growing user base by upgrading server infrastructure

Working of Web Application :

- Users sends request to a web server using either a web browser or application's user interface and internet connection is required to secure seamless access
- Web server forwards the request to the corresponding web application server
- The web application server performs the required task and generates the appropriate results
- Results will be sent back from the web application server to the web server
- Web Server responds to the client and displays the requested information to the user

Q3) What is a web server ?

- A web server stores and delivers web content to users over the internet using HTTP or HTTPS
- An HTTP server handles HTTP requests and responses which is a core component of the client server model handling requests from browsers and returning resources required
- Web servers use the HTTP protocol to manage communication on the web
- A browser sends an HTTP request to the server when enter website URL where server responds by delivering the requested web page for the browser to display

Types Of Web Server Software :

Apache HTTP Server :

- It is a free and developed as open source and works nearly on OS and most used HTTP server globally
- Supports dynamic content well and integrate scripting languages like PHP

Apache Tomcat :

- It is a open source and free web server software from Java ecosystem
- Generates server-side web content using Java

NGINX :

- It is a free and web server which includes features like reducing load on HTTP server and improves speed

Microsoft IIS :

- It is known as Microsoft internet information services which is a server software that has web server component
- It had a built in feature of Windows Server Systems
- Now installed manually on certain windows OS and can be used on Windows Server also

Web Server Applications :

Hosting Websites :

- It is commonly used as web server to host websites making them accessible on the internet

Web Application :

- They provide infrastructure for hosting web applications enabling users to interact with software through web interface

File Sharing :

- Some web servers are used for file sharing and collaboration
- Where it allows users to upload and download file securely

Content Delivery :

- Uses web servers to distribute content like images and videos to users worldwide by reducing load time

API hosting :

- Web servers are used to host API known as application programming interface which allows application to communicate and exchange data over the internet

Steps To Use Web Server :

Choosing A Web Server Software :

- By selecting a web server software that suits your needs
- Apache and NGINX are popular depending on the requirements

Installing and Configure :

- Install the chosen web server software on your server or hosting environment
- Should configure the specify settings such as document root and security rules and virtual hosts

Uploading Your Websites :

- Upload websites files to the server which includes HTML , CSS , JavaScript and other like videos or images

Testing Your Website :

- Test your website by accessing through web browser and ensure loads correctly with working functionalities

Q4) What is HTTPS protocol & Why is it considered secured?

- HTTPS stands for Hypertext Transfer Protocol Secure and a protocol for sending data between a web browser and a website
- It is a secure variant of HTTP and is used to communicate between the user , browser and the website ensuring data transfer is encrypted for added security
- It is important because it keeps the information on websites safe from being easily viewed or stolen by anyone who might be spying on the network
- When a website uses regular HTTP , data is sent in small chunks called packets that can easily be intercepted using free software
- It makes communication especially over public WIFI very vulnerable to attacks
- HTTPS encrypts the data if someone manages to intercept the packets they will appear as random , unreadable characters
- HTTPS is encrypted in order to increase security of data transfer importantly when users transmit sensitive data such as logging into bank account , email services or health insurance provider details
- Basically any website especially those that require login credentials should use HTTPS

Q5) What is Authentication & Authorization ?

- Authentication and Authorization are two fundamental security process that work together to control access to systems and data
- Authentication confirms the user identity which proves who the user is
- Authorization controls what the verified user is allowed to do which decides what can they access

Working of Authentication and Authorization :

Authentication :

- Authentication is the process of verifying the identity of a user or system and ensures that the user is legitimate by validating credentials like password , OTP , Biometrics

Working of Authentication :

- Users enters credentials like password , OTP , Biometric
- The system verifies the credentials and if valid user is successfully authentication

Authorization :

- Authorization determines the access rights and permissions of a authenticated user
- It decides what resource the user can access and what access and what actions they are allowed to perform

Working of Authorization :

- The system checks the user roles or permissions and grants or denies access to resources
- It ensures the user can perform only allowed actions

Comparison :

Authentication :

- It verifies who the user is
- It is performed before authorization
- It requires login credentials like username , password , biometric
- It determines if the user is valid
- It uses user id tokens
- The credentials can be changed by the user
- The entering credentials are visible to the user
- Password , OTP , Fingerprint , Face Recognition are the examples

Authorization :

- It determines what the user can access
- It happens after authentication
- It requires user roles , privileges or access levels
- It determines what permission the valid user has
- It uses user access tokens
- The permissions can only be changed by system owner
- This is not visible to the user and the process are handled in background
- Admin rights , read write access , role based permissions are examples

Types of Authentication :

Passwords :

- These are old school and are less secure but are still commonly used in almost all authentication schemes

One Time Password :

- Known as OTP where it grants a user or device one time access to system
- Authentication systems can deliver an OTP via email , text message or physical access token

Password less Authentication :

- Basically a variation of MFA which uses two authentication factors including fingerprint , magic links to authenticate users without password

Multi Factor Authentication :

- Basically known as MFA which combines multiple forms of authentication to improve security

Biometric Authentication :

- Uses an inherence factor such as user’s face or fingerprint to identify them

Types of Authorization :

Role Based Authorization :

- Known as RBA and access is given based on your role
- The admin can add users , delete data and manager can view reports and the user can view their own data
- Most common in companies

Permission Based Authorization :

- The access is given based on specific permissions and not roles
- There are permissions such as reading , write , delete the file
- It has more control than roles

Attribute Based Authorization :

- Known as RBA where the access depends on conditions or attributes
- If conditions are matched then access allowed
- These are used in large enterprises and condition such as user department is connected to IT and location is connected and matched to India and time matching and connecting Office Hours

Token Based Authorization :

- The access is checked using a token which is a digital key such as OAuth token
- These are very common in APIs and cloud apps

OAuth Authorization :

- We can login using another service without sharing password
- Like logging with Google and GitHub and these are used in modern apps and API’s

Real Life Examples : Authentication & Authorization :

Authentication in Health Care Systems :

- This method used to protect health care systems such as electronic health records and including credentials as username and password , smart card , biometric authentication and MFA
- This method is used in the platform often used to provide fast and secure access to electronic health system
- While health care provider staff uses passwords , biometric

Authorization in Health Care Systems :

- Tools such as RBA & ABA (Role Based Authorization & Attribute Based Authorization)
- These tools used to ensure that only authorized users can access physical and electronic health records or information
- Doctors are granted access to all patient records while receptionists access is limited to basic contact and services

Choosing The Right Method For Your System :

- Selecting appropriate authentication and authorization methods for your system is important and crucial

Based On Security Requirements :

- How sensitive is your data

Based On User Base :

- Considering the technical proficiency of your users
- Complex authentication methods might frustrate less tech savvy users

Based On Scalability :

- Will our system need to accommodate growing number of users or roles

Based On Compliance :

- Are the industry regulations or standards you need to meet

Based On Integration :

- How well will the chosen methods integrate with the existing systems

Q6) How does social login like logging into some applications via Google or Facebook ?

- Social login is single sign on for end users . Using existing login information from a social network provider like Facebook or Google
- The user can sign into a third party website instead of creating new account specifically for that website
- This simplifies registrations and login for end user

Working of Social Login :

- The user enters your application and selects that desired social network provider
- A login request is send to the social network provider and once the social network provider confirms the user identify , a current user will get access to your application
- Ag new user will be registered as a new user and then logged into the application

Advantages of Social Login :

- Social login can increase successful login and sign up rates especially if we choose providers that match your audience
- It is powerful on mobile , where smaller screens make traditional logins a hassle
- Social login reduces password fatigue by not needing to remember or reset another password
- Reduces fewer fake accounts with real accounts are tied to existing identities
- Lower maintenance costs helps developers not need to manage password infrastructure
- Social login provider is used mainly for Google for social authentication
- Most used authentication method used is password followed by social login , OTP and last magic links

Popular Providers for Social Login :

- Google social login is widely used because of its extensive reach and outstanding security features
- Facebook login service provides access to broad user base and allows app to gather additional data like user interests and social connection with user permissions
- Apple sign in is focused on privacy and security and it gives users the option to hide their email id and creates unique app specific id
- Twitter known as X social login is useful for apps targeting an audience that frequently engages with X and enables authentication along with access to basic profile information
- LinkedIn social login is ideal for professional or job related apps and allows to user professional profile data

Q7) What is synchronous and asynchronous communication between web application ?

Synchronous Communication

- It is a type of interaction in with the sender sends a request and waits for an immediate response before proceeding
- Based on system design the requesting component like for service must pause its execution until the receiving component processes the request and returns a result
- Gives immediate feedback and interaction and encourages clearer and faster decision making
- It requires all the parties to be available at the same time which can lead to interruptions and time consuming
- A phone call on face to face communication is an example

Asynchronous Communication

- It is a method which sender sends request and continues its execution without waiting for immediate response
- In system design it allows the requesting component like for service to proceed with other tasks while the receiving component processes the request independently
- Once the response is ready , it can be delivered via call back
- Provides flexibility where people can respond at their convenience
- Reduces pressure for quick replies , allowing more thoughtful responses
- Leads to fewer slower communication process and lead to misunderstanding or delays in decision making

Synchronous Communication & Asynchronous Communication used between web applications :

Synchronous Communication :

- Synchronous communication is called blocking where one application sends a request and waits until the other application replies
- So Web Application (A) sends a request to Web Application (B)
- Web Application (A) waits and Web Application (B) process the request
- Web Application (B) sends the response and Web Application (A) continues
- In a phone call we call someone and we wait and they answer and we both talk and call ends
- Therefore a browser calling an API using HTTP and login is requested and fetches user profile and checks product price

Asynchronous Communication :

- Asynchronous Communication is called non-blocking where application sends a request and does not wait and it continues doing other work
- Web Application (A) sends message to Web Application (B)
- Web Application (A) continues working and Web Application (B) processes the message later
- Result is sent back via callback or notification

Real Web Application Example :

A E-Commerce Website : (Synchronous Process) :

- User login in and user views product and user adds item to cart

A E-Commerce Website : (Asynchronous) :

- Order is waiting on confirmation in email
- They payment is waiting to process and waiting for inventory update and waits and gives SMS or Notification

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TITLE : Research Q_A_Part_2

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Q1)What is frontend and backend ?

- In web development , frontend & backend are important for understanding how websites and web application works
- Frontend and backend are two components make up the core of any modern web application each serving a unique purpose

Frontend :

- What users see and interact with on a website like layout , buttons and text
- It is the port of the websites or app users directly interact with. It includes the layout , design and interactive elements , creating a smooth and user friendly experience
- Also called the client-side because it runs in the users browser
- Focuses on designing the look of a website , including layout , colours , interactive elements using User Interface Design known as UI
- Ensures that websites work well on various devices and screen sizes providing seamless experience on desktops , tablet and smartphones by the responsiveness
- Uses interactivity to implement interactive features like buttons , forms , animation , dynamic content that engage users
- Uses cross-browser- compatibility ensures that the website functions correctly across different web browsers such as Chrome , Firefox , Safari

Backend :

- Is the part that works behind the scenes handling tasks like storing data and processing requests
- Is the server side of web development which handles and processes data and manage storage and restore
- Ensures the frontend gets the information it needs and while users interact with the frontend , the backend works behind the scenes to manage business logic and data operations
- Uses server-side logic to handle the server-side operations including processing data , managing application logic and handling user requests
- Uses database management to manage and interact with database like MYSQL, MongoDB to store , retrieve and update data based on user interactions
- Uses API Development to create and manage API which is the application programming interface which allows communication between frontend and backend
- Ensures security of data by implementing authorization and authentication , encryption and protection against attack like SQL injection

Frontend Technologies Usage :

- HTML known as Hypertext Markup Language is a standard language for creating the structure of web pages . Also defines elements such as headings , paragraphs , links and images
- CSS known as cascading style sheets used for styling HTML elements and controls the layout , colours , fonts and responsiveness of a webpage
- JavaScript is a programming language that adds interactivity to the web and used to create dynamic content such as forms , animations , interactive maps

Backend Technologies Usage :

- To handle server-side-logic and data management
- Programming languages are used such as PHP , Python , Java and C#
- Database such as MYSQL , MongoDB interact with database to store and restore data
- Web servers such as Apache , NGINX , MS IIS used to handle HTTP requests and deliver content to users

- Backend libraries such as Pandas and Backend Framework such as Python (Djano) , Ruby , Node.js , Java (Spring Boot)

Q2) Three Tier Architecture in Web Application :

- It is the golden rule of building modern web applications
- Instead of stuffing everything into one giant pile of code , we break the application into three distinct layers
- This makes web application easier to update faster to scale and secure more

The Presentation TIER (The Frontend) :

- This is the top level and the part you see actually and touch
- It is the user interface UI that runs in web browser application
- It helps to show information to the user and collect their input
- And HTML , CSS , JavaScript and framework like React , VUE , Angular are the technologies

The Application TIER (Logic or Backend) :

- This is the brain of the application and does everything as middle man and takes the data from the user process it and decides what to do next
- It also handles data processing , calculation and rules and the technology used are Python , Java , Node.js , PHP or C#

The Data TIER (The Database) :

- This is where the information is stored permanently
- The application tier talks to this layer to save or restore data
- It also stores, manage and protect data and technologies used are MYSQL , MongoDB

Therefore , We can use 3 tier instead of putting it all together by :

First is Scalability :

- We can upgrade the server without having to change the frontend

Second is Security :

- The user never touches the database directly
- The application tier acts as a security guard

Third is Reliability :

- If the frontend crashes , our data remains safe and sound in the database

Fourth is Maintenance :

- Different teams can work together on different layers at the same time

Working of 3 Tier Web Architecture for Web Applications : Real Life Example:

Presentation Tier :

- This is you we type buy shoes and click the button

Application Tier :

- The server which receives the request and checks if you have enough money in your account ad see if the shoes are in stock

Database Tier :

- Confirms the stock level is “1” and updates it to “0” once you buy them

Application Tier :

- The result which sends a success message back to screen

Q3) Three Tier Architecture in Web Development ?

Presentation Tier (Frontend / Client Layer) :

- This is what the user sees and interacts with
- It displays UI like web pages , button , forms
- It takes user input and send request to backend and show responses from backend
- HTML , CSS , JavaScript , React , Angular , Vue and Google Chrome as browser and mobile apps are some of the examples
- User opens a login page and enters username and password, in this no business logic just only UI request is sending

Application Tier (Business Logic / Backend) :

- The brain of the application where process user requests and apply business rules by authentication and authorization and validating the data by communicate with database
- Java (Spring Boot) : Python (Django) : Node.js : PHP are some of the examples
- If we check username and password are correct and decide whether user can login by fetching or update data from the database
- In this there is no UI here and no direct user interaction

Database Tier (Data Layer) :

- This stores and manages data to store data securely and restore data by updating and delete data to maintain data integrity
- MYSQL , SQL Server , MongoDB , DynamoDB are some examples
- We store user details and fetch user record during login , Here no business logic and here only data is operational

Real Examples : Login Process : Step Flow :

Presentation Tier :

- User enters login details

Application Tier :

- The request goes to backend API

Database Tier :

- The backend checks credentials in DB

Application Tier :

- Database sends result back

Presentation Tier :

- Backend sends success or failure

Example in AWS :

In Presentation Tier : S3 , Cloud Front , EC2

In Application Tier : EC2 , ECS , EKS , Lambda

In Database Tier : RDS , Dynamo DB

(The above are the AWS services with par with the TIERS)

Q4) What is SSL Encryption ?

- It is a secure socket layer known as SSL is an internet security protocol that encrypts data to ensure secure communication between devices over a network
- SSL provides privacy , authentication and data integrity for online communication
- SSL is the predecessor of TLS such as transport layer security which is now the standard protocol for secure communication on the internet
- SSL ensures secure communication through three main mechanisms such as :

Encryption :

- Data is transmitted over the network is encrypted preventing unauthorized parties from reading it.
- If intercepted and encrypted data appears as an unreadable jumble of characters

Authentication :

- SSL uses handshake process to authenticate both the client and server ensuring each party is legitimate

Data Integrity :

- SSL digitally signs transmitted data to detect any tampering ensuring that the data received is exactly what was sent
- SSL is important because it encrypts sensitive information such as login credentials , financial transactions and personal data
- Also authenticating web servers to prevent users from connecting to fraud websites
- Also ensuring data integrity so transmitted information cannot be modified during transit

SSL Protocols :

SSL Record Protocol :

- Provides confidentiality and message integrity
- Application data is divided into fragment optionally compressed and appended with a message authentication code (MAC)
- Data is then encrypted and transmitted with an SSL header

Handshake Protocol :

- Establishes SSL session and authenticates clients and servers

Four Phases :

- Client and server exchange packets and protocol versions and cipher suites
- Server sends its certificate and server key information
- Client responds with its certificate and key exchange
- Changing cipher spec finalizes the handshake by activating secure communication

Alert Protocol :

- Communicate SSL related warnings or errors
- Warning alerts (Level 1) : Its non- critical issues : Expired or unsupported certificates
- Fatal alerts (Level 2) : Its critical errors : Such as handshake failures and Bad record of parameters terminates the connection

Versions of SSL / TLS :

SSL 1 :

- First version and not secure

SSL 2 :

- Created in 1995
- First public release

SSL 3 :

- Created in 1996
- They improved the security

TLS .1.0 :

- Created in 1999
- Successor to SSL 3.0

TLS .1.1 :

- Created in 2006
- Improved encryption and security

TLS.1.2 :

- Created in 2008
- Widely adopted and strong encryption

TLS .1.3 :

- Created in 2018
- Modern efficient and secure protocol

SSL Certificates : (Features)

- Digital certificates issued by trusted certificate authorities to secure and verify websites
- Protects sensitive information during transmission by encryption
- Confirms the identity of the websites or service by authentication
- Ensures transmitted data is not altered by integrity
- Prevents denial of transmitted messages
- Facilitates secure key exchange through public key
- Secure sessions using cryptography after interruptions

Types of SSL Certificate :

Single Domain :

- Secures one domain

Wild Card :

- Secures one domain and all subdomains

Multi Domain :

- Secures multiple domains in one certificate

Validation Levels :

Domain :

- Confirms domain ownership

Organization :

- Confirms organization identity

Q5) What are the HTTP methods ?

- HTTP known as Hypertext Transfer Protocol
- Specifies a collection of request methods to specify what actions is to be performed on particular resource
- HTTP request methods are GET , POST , PUT , PATCH , Delete
- It is equal to CRUD Operations
- Requests are the messages sent by the client to request data from the server or to perform some actions

GET :

- Requests used to read or restore data from web server
- Returns as an HTTP status code of 200 (Ok)
- Data is successfully restored from the server

POST :

- Request used to send data like file , form data to the server
- Successful creation returns HTTP status code of 201

PUT :

- Request is used to modify the data on the server
- Replaces the entire content at a particular location
- No resources matched the request will generate one

PATCH :

- Similar to PUT request
- Modifies a part of the data
- It replaces the content that you want o delete and replace

DELETE:

- Request used to delete the data on the server at a specified location

Q6) What are the HTTP status codes ?

- These codes are like short hand notes a server sends back to your browser
- Tell exactly what happened when the browser asked to see a page whether things went perfectly

1xx : International :

- The server has received the request and is continuing the process
- 100 Continue: Everything is fine so far; keep going with the request.

2xx : Success :

- This means the request was received, understood, and accepted.
- 200 OK: The standard "Success" code. You got the webpage you wanted.
- 201 Created: The request was successful and a new resource (like a new user account) was created.

3xx: Redirection :

The resource you're looking for has moved elsewhere.

- 301 Moved Permanently: The page lives at a new URL now. Your browser will automatically go there.
- 304 Not Modified: "You already have the latest version in your cache, so I'm not sending it again." (Saves data)

4xx: Client Error :

These are "your fault" (or at least, the fault of the browser/user).

- 400 Bad Request: The server didn't understand what you asked for.
- 401 Unauthorized: You need to log in to see this.
- 403 Forbidden: The server understands you, but it refuses to let you in. No entry!
- 404 Not Found: The classic. The server can't find the page you requested. It might have been deleted or the URL is typed wrong.

5xx: Server Error :

These are the server's fault. It knows it messed up.

- 500 Internal Server Error: A generic "catch-all" error. Something broke behind the scenes.
- 502 Bad Gateway: One server got an invalid response from another server it was trying to talk to.
- 503 Service Unavailable: The server is overloaded or down for maintenance. "Come **back later.**"

Q7) CRUD operations :

- CRUD stands for Create, Read, Update, and Delete

Create (The "C")

This is the process of adding new data to your system.

- Action: Generating a new record.
- Real-world Example: Signing up for a new account or posting a tweet.
- HTTP Method: POST
- SQL Command: INSERT

Read (The "R")

This is how you retrieve or view existing data without changing it.

- Action: Fetching data to display it on a screen.
- Real-world Example: Scrolling through your feed or searching for a specific product on Amazon.
- HTTP Method: GET
- SQL Command: SELECT

Update (The "U")

This is used to modify data that already exists in the database.

- Action: Editing an existing record.
- Real-world Example: Changing your profile picture or editing a typo in a blog post.
- HTTP Method: PUT (replaces the whole record) or PATCH (updates specific parts).
- SQL Command: UPDATE

Delete (The "D")

This is the process of removing data from the system.

- Action: Destroying a record so it no longer exists.
- Real-world Example: Deleting an old email or deactivating an account.
- HTTP Method: DELETE
- SQL Command: DELETE